R Notebook

```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(tidyr)
library(stringr)
library(ggplot2)
library(RColorBrewer)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats   1.0.0     v readr     2.1.5
## v lubridate 1.9.4     v tibble    3.3.0
## v purrr     1.0.4

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(knitr)
library(kableExtra)
```

```
##
## Attaching package: 'kableExtra'
##
## The following object is masked from 'package:dplyr':
##
##     group_rows
```

```r
library(rstatix)
```

```
##
## Attaching package: 'rstatix'
##
## The following object is masked from 'package:stats':
##
##     filter
```

```
library(coin)
```

```
## Loading required package: survival
##
## Attaching package: 'coin'
##
## The following objects are masked from 'package:rstatix':
##
##     chisq_test, friedman_test, kruskal_test, sign_test, wilcox_test
```

```
library(patchwork)
library(ggpubr)
library(viridis)
```

```
## Loading required package: viridisLite
```

```
source("/Users/mattparker/Dropbox/Workspace/moops/manuscript1/helper_utils.r")
knitr::opts_chunk$set(dev = "cairo_pdf")
```

```
# List all CSV files in the directory
csv_files <- list.files(path = "/Users/mattparker/Dropbox/Workspace/moops/manuscript1/data/experiment_al
seq_metrics = read.csv("/Users/mattparker/Dropbox/Workspace/moops/manuscript1/data/sequencing_summary_m

# Process all CSV files and combine them
all_data <- bind_rows(lapply(csv_files, process_csv_file))

# separate slurry from pond samples
pond_samples = c("s9","s10","s19","s20")

# Create a metadata dataframe to merge with taxon data
sample_metadata = seq_metrics %>%
  select(sample_name, water_control, infection_class) %>%
  mutate(
  # Library prep: MSSPE if ends with "_M"
  library_prep = if_else(str_ends(sample_name, "_M"), "MSSPE", "NGS"),

  # Remove the _M suffix to get the core sample identifier
  core = str_remove(sample_name, "_M$"),
  # Detect experiment 3 first (Cxxx_LPx)
  is_exp3 = str_detect(core, "C\\d+.*LP\\d+"),

  # Base: first sample ID (e.g. S10, PC1)
  base = str_extract(core, "^[A-Za-z]+\\d+"),

  # Append -2 or -3 only if not exp3
```

```r
  suffix = case_when(
    !is_exp3 & str_detect(core, "_2$") ~ "-2", #
    !is_exp3 & str_detect(core, "_3$") ~ "-3", #
    TRUE ~ ""
  ),

  # lowercase base sample name
  base_sample = tolower(base),

  # mark which experiment number
  experiment = case_when(
    is_exp3 ~ 3,
    str_detect(core, "_3$") ~ 2,
    TRUE ~ 1
  ),
  # group slurry vs environmental samples
    sample_type = if_else(base_sample %in% pond_samples, "environmental", "slurry")

) %>%
select(sample_name, base_sample, library_prep, experiment, water_control, infection_class, sample_type)

# Merge the taxonomy data with metadata
merged_data <- all_data %>%
  left_join(sample_metadata, by = "sample_name")

# get hpv data from experiment 1 & 2
hpv_data = merged_data %>%
  filter(experiment == 1 | experiment == 2) %>%
  filter(str_detect(name, regex("Alphapapillomavirus", ignore_case = TRUE)))

# kobuvirus data from experiment 1 to compare to experiment 3
pkv_data = merged_data %>%
  filter(experiment == 1) %>%
  filter(str_detect(name, regex("Aichivirus", ignore_case = TRUE)))

# remove MSSPE libraries that are also experiment 1 (after we have already pulled the hpv rows)
merged_data = merged_data %>% filter(!(experiment == 1 & library_prep == "MSSPE"))

# Filter to include only species level (tax_level == 1) to avoid duplication
all_data_species <- merged_data %>%
  filter(tax_level == 1) %>%
  filter(experiment == 2 | experiment == 3) %>%  # filter in specific experiments
  #filter(experiment == 1) %>%
  filter(water_control == "No") %>%   # filter out Negative Controls
  filter(!base_sample %in% c("pc1","pc3")) %>%  # filter out zymo fecal controls
  filter(infection_class != "Definite")  # filter out positive controls
  #filter(sample_type == "slurry") # filter to only slurry

# add hpv & kobuvirus data to the rest of the data
all_data_species = all_data_species %>%
  add_row(hpv_data) %>%
  add_row(pkv_data)
```

```r
# targets to filter to the genus level
genus_targets = c(
  "Bocaparvovirus",
  "Sapelovirus",
  "Circovirus",
  "Enterovirus",
  "Teschovirus",
  "Alphapapillomavirus",
  "Sapovirus",
  "Mamastrovirus",
  "Kobuvirus",
  "Asfivirus",
  "Pestivirus",
  "Aphthovirus",
  "Betaarterivirus",
  "Alphainfluenzavirus",
  "Henipavirus")

# assign these to group species
name_targets = c(
  "PBoV", # porcine bocavirus
  "PCV", # porcine circovirus 2
  "PKV", # porcine kobuvirus / aichivirus C
  "EV-G", # porcine enterovirus G
  "PTeV", # porcine teschovirus A
  "PAstV", # porcine astrovirus 4
  "PSapoV", # porcine sapovirus
  "PSapeV", # porcine sapelovirus
  "HPV18" # humanpapillomavirus 7/18
  #"PToV*" # porcine torovirus
)

# specific names that should group together
ent_v = "Enterovirus G"
circo_v = "Circovirus"
sapo_v = paste(c("Sapovirus","Sapporo"), collapse = "|")
tes_v = "Teschovirus"
aich_v = "Aichivirus"
astr_v = paste(c("Porcine astrovirus","Mamastrovirus 3"), collapse = "|")
hpv18 = "Alphapapillomavirus"
sapel_v = "Sapelovirus"
boc_v = paste(c("Porcine bocavirus","Bocaparvovirus"), collapse = "|")
#toro_v = "torovirus"

all_viruses <- all_data_species %>%
  filter(category == "viruses") %>%
  filter(is_phage == "false")  # filter out phages

# add name group column
all_viruses = all_viruses %>%
  mutate(name_group = case_when(
    str_detect(name, regex(hpv18, ignore_case = TRUE)) ~ "HPV18",
    str_detect(name, regex(aich_v, ignore_case = TRUE)) ~ "PKV",
```

```r
    str_detect(name, regex(tes_v, ignore_case = TRUE)) ~ "PTeV",
    str_detect(name, regex(circo_v, ignore_case = TRUE)) ~ "PCV",
    str_detect(name, regex(sapo_v, ignore_case = TRUE)) ~ "PSapeV",
    str_detect(name, regex(sapel_v, ignore_case = TRUE)) ~ "PSapoV",
    str_detect(name, regex(ent_v, ignore_case = TRUE)) ~ "EV-G",
    str_detect(name, regex(boc_v, ignore_case = TRUE)) ~ "PBoV",
    str_detect(name, regex(astr_v, ignore_case = TRUE)) ~ "PAstV",
    #str_detect(name, regex(toro_v, ignore_case = TRUE)) ~ "PToV*",
    TRUE ~ name  # Keep original name if no pattern matches
  ))

# Add genus columns to the dataframe
#genus_data <- get_genus_info_batch(all_viruses$tax_id, batch_size = 100)

# Combine with original dataframe
#all_viruses <- cbind(all_viruses, genus_data)

# for 0 rpm values
eps <- 0.1

# filter to only the target viral species or genus
only_viral_targets = all_viruses %>%
  group_by(base_sample, library_prep, name_group) %>%
  filter(str_detect(name_group, regex(paste0("\\b(", paste(name_targets, collapse="|"), ")\\b"), ignore_
  summarize(rpm_mean = mean(nt_rpm, na.rm=TRUE), .groups="drop") %>%
  pivot_wider(
    names_from = library_prep,
    values_from = rpm_mean
  ) %>%
  mutate(
    MSSPE = replace_na(MSSPE, 0), # set NAs to 0 for non detections
    NGS   = replace_na(NGS, 0), # set NAs to 0 for non detections
    mean_rpm_enrichment = (`MSSPE` + eps)/(`NGS` + eps),
    log2FC = log2(mean_rpm_enrichment),
    name_group = fct_reorder(name_group, log2FC, .fun = mean, .desc = TRUE)
  ) %>%
  arrange(desc(log2FC))

# calculate average enrichment for all targets
mean_enrich_overall = only_viral_targets %>%
  filter(name_group != "PToV*") %>%
  summarise(mean_log2FC = mean(log2FC), median_foldchange = median(mean_rpm_enrichment))

mean_enrich_overall
```

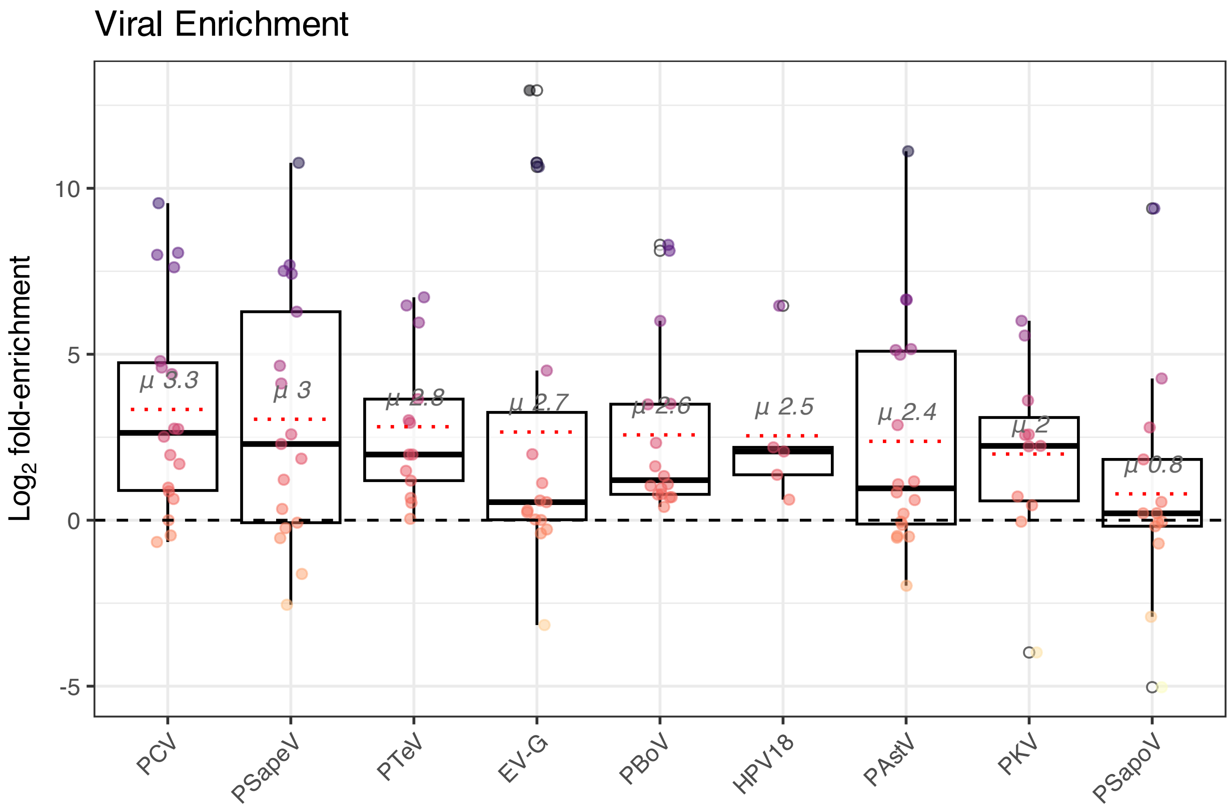
```
## # A tibble: 1 x 2
##   mean_log2FC median_foldchange
##         <dbl>             <dbl>
## 1        2.52              2.94
```

```r
# calculate mean so we can add to the boxplot
mean_df <- only_viral_targets %>%
  group_by(name_group) %>%
```

```
  summarise(mean_value = mean(log2FC, na.rm = TRUE))

# plot fold change across all samples
alltarget_boxplot <- ggplot(only_viral_targets, aes(x = name_group, y = log2FC)) +
  geom_boxplot(width = 0.8, outlier.shape = 21, outlier.fill = "white", alpha = 0.6, width = 0.6, color
  geom_jitter(width = 0.1, alpha = 0.5, size = 1.5, aes(colour = log2FC)) +
  scale_color_viridis(option = "magma", direction = -1) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  # dotted mean line
  geom_segment(
    data = mean_df,
    aes(x = as.numeric(name_group) - 0.3,
        xend = as.numeric(name_group) + 0.3,
        y = mean_value, yend = mean_value),
    inherit.aes = FALSE,
    linetype = "dotted", color = "red", linewidth = 0.6
  ) +
  geom_text(data = mean_df, aes(x = name_group, y = mean_value, label = paste0("  ", round(mean_value, 1
  scale_x_discrete() +  # adds space between boxes for labels
  labs(
    title = "Viral Enrichment",
    x = "",
    y = expression(Log[2]*" fold-enrichment"),
  ) +
  theme_bw() +
  theme(
    legend.position = "none",
    axis.text.x = element_text(angle = 45, hjust = 1),
  )
```

```
## Warning: Duplicated aesthetics after name standardisation: width
```

```
alltarget_boxplot
```



Viral Enrichment w/o primers from experiment 1

```
merged_data_exp1 <- all_data %>%
  left_join(sample_metadata, by = "sample_name")

# Filter to include only species level (tax_level == 1) to avoid duplication
all_data_species_exp1 <- merged_data_exp1 %>%
  filter(tax_level == 1) %>%
  filter(experiment == 1) %>% #only experiment 1
  filter(water_control == "No") %>%   # filter out Negative Controls
  filter(!base_sample %in% c("pc1","pc3")) %>%  # filter out zymo fecal controls
  filter(infection_class != "Definite")

all_viruses_exp1 <- all_data_species_exp1 %>%
  filter(category == "viruses") %>%
  filter(is_phage == "false")  # filter out phages

# add name group column
all_viruses_exp1 = all_viruses_exp1 %>%
  mutate(name_group = case_when(
    str_detect(name, regex(aich_v, ignore_case = TRUE)) ~ "PKV",
    str_detect(name, regex(tes_v, ignore_case = TRUE)) ~ "PTeV",
    str_detect(name, regex(circo_v, ignore_case = TRUE)) ~ "PCV",
    str_detect(name, regex(sapo_v, ignore_case = TRUE)) ~ "PSapeV",
    str_detect(name, regex(sapel_v, ignore_case = TRUE)) ~ "PSapoV",
    str_detect(name, regex(ent_v, ignore_case = TRUE)) ~ "EV-G",
    str_detect(name, regex(boc_v, ignore_case = TRUE)) ~ "PBoV",
    str_detect(name, regex(astr_v, ignore_case = TRUE)) ~ "PAstV",
    TRUE ~ name  # Keep original name if no pattern matches
```

```r
  ))

# for 0 rpm values
eps <- 0.1

# filter to only the target viral species or genus
only_viral_targets_exp1 = all_viruses_exp1 %>%
  group_by(base_sample, library_prep, name_group) %>%
  filter(str_detect(name_group, regex(paste0("\\b(", paste(name_targets, collapse="|"), ")\\b"), ignore_
  summarize(rpm_mean = mean(nt_rpm, na.rm=TRUE), .groups="drop") %>%
  pivot_wider(
    names_from = library_prep,
    values_from = rpm_mean
  ) %>%
  mutate(
    MSSPE = replace_na(MSSPE, 0), # set NAs to 0 for non detections
    NGS   = replace_na(NGS, 0), # set NAs to 0 for non detections
    mean_rpm_enrichment = (`MSSPE` + eps)/(`NGS` + eps),
    log2FC = log2(mean_rpm_enrichment),
    name_group = fct_reorder(name_group, log2FC, .fun = mean, .desc = TRUE)
  ) %>%
  arrange(desc(log2FC))

# calculate average enrichment for all targets
mean_enrich_overall_exp1 = only_viral_targets_exp1 %>%
  filter(name_group != "PToV*") %>%
  summarise(mean_log2FC = mean(log2FC), median_foldchange = median(mean_rpm_enrichment))

mean_enrich_overall_exp1
```

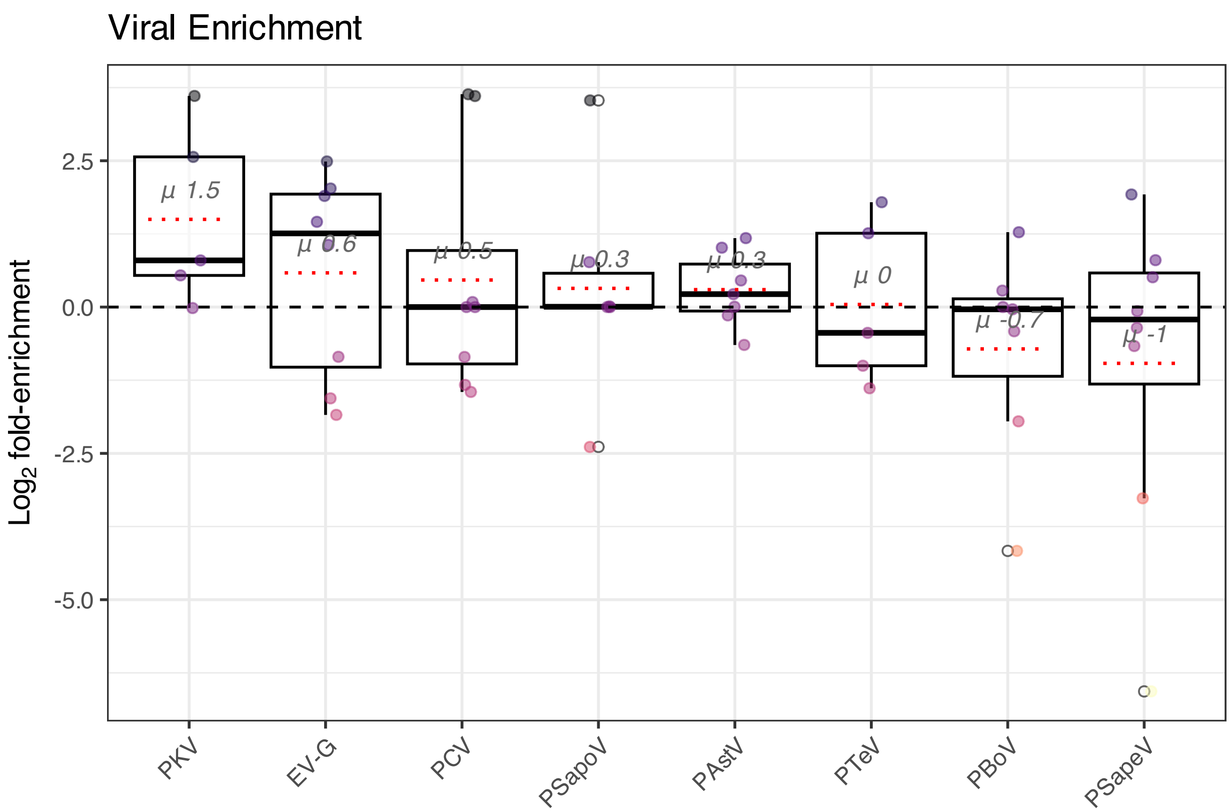
```
## # A tibble: 1 x 2
##   mean_log2FC median_foldchange
##         <dbl>             <dbl>
## 1       0.137              1.00
```

```r
# calculate mean so we can add to the boxplot
mean_df_exp1 <- only_viral_targets_exp1 %>%
  group_by(name_group) %>%
  summarise(mean_value = mean(log2FC, na.rm = TRUE))

# plot fold change across all samples
alltarget_boxplot_exp1 <- ggplot(only_viral_targets_exp1, aes(x = name_group, y = log2FC)) +
  geom_boxplot(width = 0.8, outlier.shape = 21, outlier.fill = "white", alpha = 0.6, width = 0.6, color
  geom_jitter(width = 0.1, alpha = 0.5, size = 1.5, aes(colour = log2FC)) +
  scale_color_viridis(option = "magma", direction = -1) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  # dotted mean line
  geom_segment(
    data = mean_df_exp1,
    aes(x = as.numeric(name_group) - 0.3,
        xend = as.numeric(name_group) + 0.3,
        y = mean_value, yend = mean_value),
    inherit.aes = FALSE,
```

```
    linetype = "dotted", color = "red", linewidth = 0.6
  ) +
  geom_text(data = mean_df_exp1, aes(x = name_group, y = mean_value, label = paste0("  ", round(mean_val
  scale_x_discrete() +  # adds space between boxes for labels
  labs(
    title = "Viral Enrichment",
    x = "",
    y = expression(Log[2]*" fold-enrichment"),
  ) +
  theme_bw() +
  theme(
    legend.position = "none",
    axis.text.x = element_text(angle = 45, hjust = 1),
  )
```

```
## Warning: Duplicated aesthetics after name standardisation: width
```

```
alltarget_boxplot_exp1
```



Final rPM Enrichment Tables for Manuscript

```
cat("\\captionsetup{labelformat=empty}")
```

```
# final table for each viral target
viral_target_enrichment_summary = only_viral_targets %>%
  group_by(name_group) %>%
  summarise(
```

```r
    mean_rpm_ngs = round(mean(NGS, na.rm = TRUE),2),
    mean_rpm_msspe = round(mean(MSSPE, na.rm = TRUE),2),
    median_enrichment = round(median(mean_rpm_enrichment, na.rm = TRUE),2),
    log_fold_enrichment = round(mean(log2FC, na.rm = TRUE),2),
    iqr = paste0(
      round(quantile(mean_rpm_enrichment, 0.25, na.rm = TRUE), 1),
      "-",
      round(quantile(mean_rpm_enrichment, 0.75, na.rm = TRUE), 1)
    ),
    n_pairs = n(),
    p_value = {
      # ensure both vectors are aligned and remove any NA pairs
      paired_data <- na.omit(data.frame(ngs = NGS, msspe = MSSPE))

      if(nrow(paired_data) > 1) {
        round(wilcox.test(paired_data$ngs,
                    paired_data$msspe,
                    alternative = "two.sided",
                    paired = TRUE)$p.value, 3)
      } else {
        NA_real_
      }
    },
  effect_r = {
      paired_data <- na.omit(data.frame(ngs = NGS, msspe = MSSPE))
      if(nrow(paired_data) > 1) {
        long_df <- paired_data %>%
          pivot_longer(cols = c(ngs, msspe), names_to = "prep", values_to = "value") %>%
          mutate(prep = factor(prep, levels = c("ngs","msspe")))
        round(wilcox_effsize(long_df, value ~ prep, paired = TRUE)$effsize, 3)
      } else NA_real_
    }
  ) %>%
  arrange(desc(median_enrichment))
```

```
## Warning: There was 1 warning in `summarise()`.
## i In argument: `p_value = { ... }`.
## i In group 1: `name_group = PCV`.
## Caused by warning in `wilcox.test.default()`:
## ! cannot compute exact p-value with zeroes
```

```r
# Make table pretty
viral_target_enrichment_summary %>%
  rename(
    "Virus" = name_group,
    "rPM (NGS)" = mean_rpm_ngs,
    "rPM (MSSPE)" = mean_rpm_msspe,
    "Median Enrichment" = median_enrichment,
    "Log Fold Change" = log_fold_enrichment,
    "IQR" = iqr,
    "p-value" = p_value,
    "effect size (r)" = effect_r
  ) %>%
```

Enrichment across all paired samples per viral target

Virus	rPM (NGS)	rPM (MSSPE)	Median Enrichment	Log Fold Change	IQR	p-value	effect size (r)
PCV	9.32	27.93	6.21	3.34	1.9–26.9	0.005	0.688
PSapeV	12.29	29.35	4.91	3.04	0.9–77.8	0.064	0.454
PKV	36.27	50.56	4.72	1.99	1.5–9.1	0.042	0.617
HPV18	2.33	8.25	4.20	2.54	2.6–4.6	0.062	0.905
PTeV	4.11	9.78	3.94	2.82	2.3–12.5	0.000	0.882
PBoV	65.17	128.34	2.32	2.57	1.7–11.3	0.000	0.879
PAstV	14.68	30.61	1.95	2.38	0.9–34.1	0.167	0.334
EV-G	10.45	87.59	1.46	2.66	1–13.4	0.012	0.631
PSapoV	6.56	12.09	1.15	0.79	0.9–3.6	0.414	0.242

```
  select(-n_pairs) %>%
  kable(
    format = "latex",  # or "latex" if using LaTeX
    caption = "Enrichment across all paired samples per viral target",
    digits = 3,
    format.args = list(big.mark = ",")
  ) %>%
  kable_styling(
    full_width = TRUE,
    position = "center",
  )
```

Proportion of samples with viral targets

```
count_target_viruses = all_viruses %>%
  filter(!str_detect(sample_name, "PC")) %>%
  group_by(sample_name, name_group) %>%
  filter(str_detect(name_group, regex(paste0("\\b(", paste(name_targets, collapse="|"), ")\\b"), ignore_
  summarize(count = sum(n(), na.rm=TRUE), .groups="drop")

virus_sample_per_targets = count_target_viruses %>%
  group_by(name_group) %>%
  summarize(
    perc_of_samples = n_distinct(sample_name) / n_distinct(.$sample_name)
    ) %>%
  arrange(desc(perc_of_samples))
```

GENOME COVERAGE

```
cat("\\captionsetup{labelformat=empty}")
```

```
# Folder containing coverage CSVs
cov_dir <- "/Users/mattparker/Dropbox/Workspace/moops/manuscript1/data/experiment_all/coverage_tables"

# Read all CSV files
cov_files <- list.files(cov_dir, pattern = "\\.csv$", full.names = TRUE)
```

```r
# Function to compute breadth and mean depth from one coverage file
calc_cov_stats <- function(file) {
  df <- read.csv(file)
  tibble(
    file = basename(file),
    virus = str_replace(file, "_s.*|_pc.*", ""),
    base_sample = str_remove(str_extract(basename(file), "(?<=_)(s\\d+|pc\\d+)"), "_"),
    prep = ifelse(str_detect(file, "_m"), "MSSPE", "NGS"),
    sample_type = if_else(base_sample %in% pond_samples, "environmental", "slurry"),
    ref_length = nrow(df),
    breadth_1x = mean(df$Coverage > 0) * 100,
    mean_depth = mean(df$Coverage)
  )
}

# Calculate coverage for all files
cov_summary <- map_dfr(cov_files, calc_cov_stats)

# Pivot wider for paired comparison
paired_tbl <- cov_summary %>%
  #filter(sample_type == "slurry") %>% # filtering by sample type
  select(virus, base_sample, prep, breadth_1x, mean_depth) %>%
  pivot_wider(
    names_from = prep,
    values_from = c(breadth_1x, mean_depth),
    names_glue = "{.value}_{prep}"
  ) %>%
  mutate(
    enrichment_breadth = breadth_1x_MSSPE - breadth_1x_NGS,
    enrichment_depth   = mean_depth_MSSPE - mean_depth_NGS,
    virus = recode(virus,
      `astrov` = "PAstV",
      `etvg` = "EV-G",
      `hpv18` = "HPV18",
      `pbov` = "PBoV",
      `pcv` = "PCV",
      `sapel` = "PSapeV",
      `tev` = "PTeV",
      `pkv` = "PKV",
    )
  )

# Summary table (1 row per metric)
wilcox_summary = paired_tbl %>%
  replace_na(list(breadth_1x_MSSPE = 0, breadth_1x_NGS = 0, enrichment_breadth = 0)) %>% # some samples
  group_by(virus) %>%
  summarise(
    mean_ngs = round(mean(breadth_1x_NGS, na.rm = TRUE),2),
    mean_msspe = round(mean(breadth_1x_MSSPE, na.rm = TRUE),2),
    percent_enrichment = round(mean_msspe - mean_ngs,2),
    iqr = paste0(
      round(quantile(breadth_1x_MSSPE - breadth_1x_NGS, 0.25, na.rm = TRUE), 1),
      "-",
```

```
      round(quantile(breadth_1x_MSSPE - breadth_1x_NGS, 0.75, na.rm = TRUE), 1)
    ),
    n_pairs = n(),
    p_value = if (n() > 1) {
      test <- wilcox.test(
        breadth_1x_MSSPE,
        breadth_1x_NGS,
        paired = TRUE,
        alternative = "two.sided",
        exact = FALSE
      )
      round(test$p.value, 3)
    } else {
      NA_real_
    },
    effect_r = {
      paired_data <- na.omit(data.frame(ngs = breadth_1x_NGS, msspe = breadth_1x_MSSPE))
      if(nrow(paired_data) > 1) {
        long_df <- paired_data %>%
          pivot_longer(cols = c(ngs, msspe), names_to = "prep", values_to = "value") %>%
          mutate(prep = factor(prep, levels = c("ngs","msspe")))
        round(wilcox_effsize(long_df, value ~ prep, paired = TRUE)$effsize, 3)
      } else NA_real_
    },
    .groups = "drop"
  )


# Make table pretty
wilcox_summary %>%
  mutate(
    virus = recode(virus,
      `astrov` = "PAstV",
      `etvg` = "EV-G",
      `hpv18` = "HPV18",
      `pbov` = "PBoV",
      `pcv` = "PCV",
      `sapel` = "PSapeV",
      `tev` = "PTeV",
      `pkv` = "PKV",
    )
  ) %>%
  rename(
    "Virus" = virus,
    "% coverage (NGS)" = mean_ngs,
    "% coverage (MSSPE)" = mean_msspe,
    "Enrichment (%)" = percent_enrichment,
    "IQR" = iqr,
    "p-value" = p_value,
    "Effect Size (r)" = effect_r
  ) %>%
  select(-n_pairs) %>%
  kable(
```

Genome coverage enrichment across all paired samples per viral target

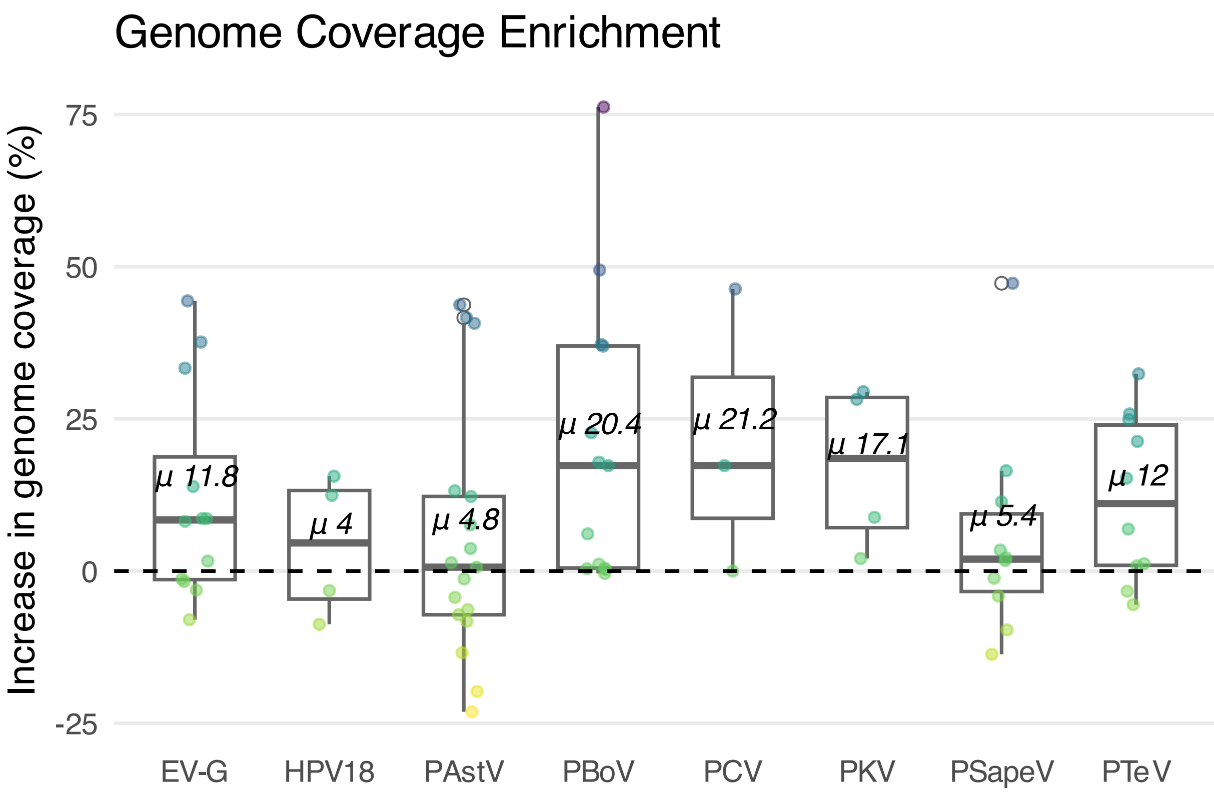
Virus	% coverage (NGS)	% coverage (MSSPE)	Enrichment (%)	IQR	p-value	Effect Size (r)
EV-G	40.96	48.62	7.66	-3.1–9.9	0.205	0.323
HPV18	15.23	19.25	4.02	-4.6–13.2	0.584	0.365
PAstV	52.43	57.14	4.71	-7–11.1	0.663	0.108
PBoV	56.90	74.51	17.61	0.4–29.8	0.006	0.719
PCV	34.17	56.66	22.49	17.6–28.2	0.059	0.861
PKV	11.44	20.60	9.16	2.3–9.4	0.019	0.757
PSapeV	32.81	36.37	3.56	-3.8–3.3	0.616	0.143
PTeV	33.49	43.40	9.91	1.2–21.3	0.025	0.630

```
    format = "latex",  # or "latex" if using LaTeX
    caption = "Genome coverage enrichment across all paired samples per viral target",
    digits = 3,
    format.args = list(big.mark = ",")
  ) %>%
  kable_styling(
    full_width = TRUE,
    position = "center",
  )
```

```
# Create the boxplot for each targeted virus
ggplot(paired_tbl, aes(x = virus, y = enrichment_breadth)) +
  geom_boxplot(width = 0.6, outlier.shape = 21, outlier.fill = "white", color = "gray40") +
  geom_jitter(width = 0.1, alpha = 0.5, size = 1.5, aes(colour = enrichment_breadth)) +
  scale_color_viridis(option = "viridis", direction = -1) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  # Annotate median on each box
  stat_summary(
    fun = mean,
    geom = "text",
    aes(label = paste0("  ", round(..y.., 1))),
    vjust = -0.5,
    size = 4,
    color = "black",
    fontface = "italic"
  ) +
  labs(
    x = "",
    y = "Increase in genome coverage (%)",
    title = "Genome Coverage Enrichment"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    legend.position = "none",
    panel.grid.major.x = element_blank(),
    panel.grid.minor = element_blank()
  )
```

```
## Warning: The dot-dot notation (`..y..`) was deprecated in ggplot2 3.4.0.
```

```
## i Please use `after_stat(y)` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

## Warning: Removed 23 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

## Warning: Removed 23 rows containing non-finite outside the scale range
## (`stat_summary()`).

## Warning: Removed 23 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```
# Overall Enrichment (non-paired)


# All viruses & enrichment
overall_tbl = all_viruses %>%
  filter(str_detect(name_group, regex(paste0("\\b(", paste(name_targets, collapse="|"), ")\\b"), ignore_
  filter(name_group != "PToV*") %>% # remove torovirus
  group_by(name_group, library_prep) %>%
  summarize(rpm_mean = mean(nt_rpm, na.rm = TRUE), .groups = "drop")

overall_tbl_summary = all_viruses %>%
  filter(str_detect(name_group, regex(paste0("\\b(", paste(name_targets, collapse="|"), ")\\b"), ignore_
  filter(name_group != "PToV*") %>% # remove torovirus
```

```r
  group_by(base_sample, name_group, library_prep) %>%
  summarize(rpm_mean = mean(nt_rpm, na.rm = TRUE), .groups = "drop") %>%
  pivot_wider(
    names_from = library_prep,
    values_from = rpm_mean
  ) %>%
  mutate(
    MSSPE = replace_na(MSSPE, 0), # set NAs to 0 for non detections
    NGS   = replace_na(NGS, 0), # set NAs to 0 for non detections
    mean_rpm_enrichment = (`MSSPE` + eps)/(`NGS` + eps),
    log2FC = log2(mean_rpm_enrichment),
    name_group = fct_reorder(name_group, log2FC, .fun = mean, .desc = TRUE)
  ) %>%
  arrange(desc(log2FC))

# Calculate average enrichment overall
median(overall_tbl_summary$log2FC)
```

```
## [1] 1.55621
```

```r
round(quantile(overall_tbl_summary$log2FC, 0.25, na.rm = TRUE), 2)
```

```
##  25%
## 0.25
```

```r
round(quantile(overall_tbl_summary$log2FC, 0.75, na.rm = TRUE), 2)
```

```
##  75%
## 4.48
```

```r
# median enrichment overall
median(overall_tbl_summary$mean_rpm_enrichment)
```

```
## [1] 2.944116
```

```r
round(quantile(overall_tbl_summary$mean_rpm_enrichment, 0.25, na.rm = TRUE), 2)
```

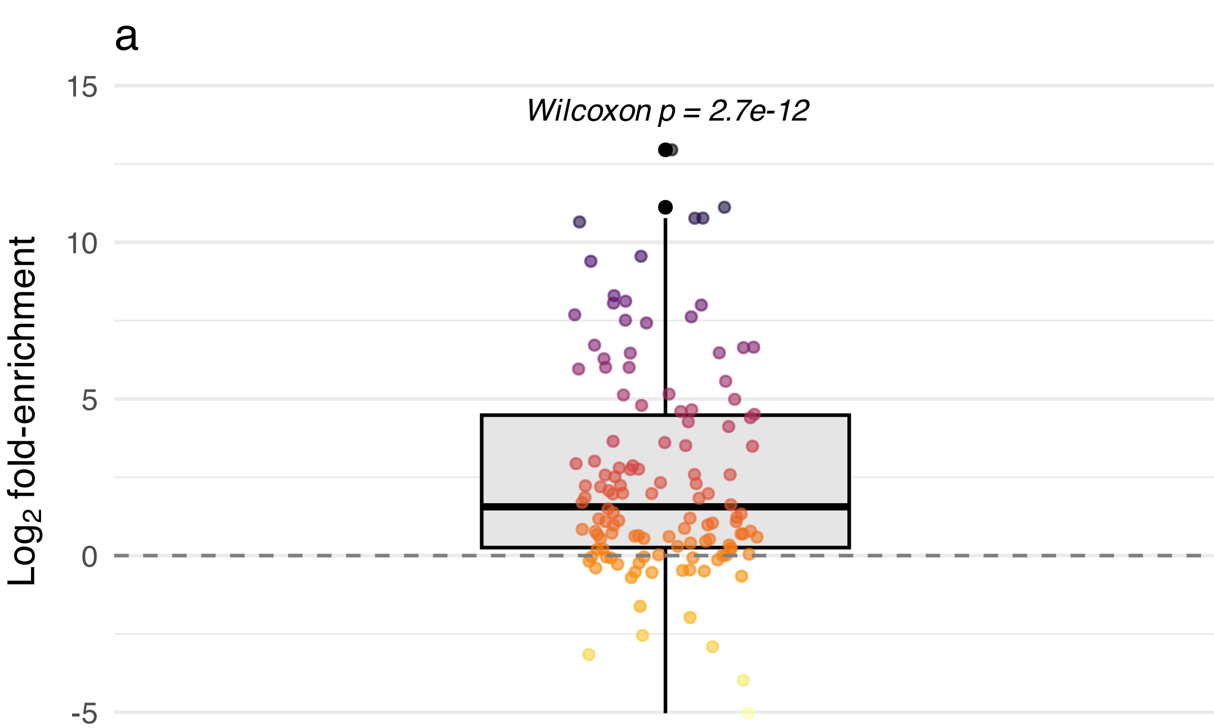
```
##  25%
## 1.19
```

```r
round(quantile(overall_tbl_summary$mean_rpm_enrichment, 0.75, na.rm = TRUE), 2)
```

```
##   75%
## 22.35
```

```r
# Paired Wilcoxon test across all samples
wilcox_res <- wilcox.test(
  overall_tbl_summary$NGS,
  overall_tbl_summary$MSSPE,
```

```
  paired = TRUE,
  alternative = "two.sided"
)

pval <- wilcox_res$p.value

log2fc_plot = ggplot(overall_tbl_summary, aes(x = "", y = log2FC)) +
  geom_boxplot(width = 0.4, fill = "gray90", color = "black") +
  geom_jitter(width = 0.1, alpha = 0.6, size = 1.5, aes(color = log2FC)) +
  scale_color_viridis(option = "inferno", direction = -1) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
  annotate("text", x = 1, y = max(overall_tbl_summary$log2FC, na.rm = TRUE) * 1.1,
           label = paste0("Wilcoxon p = ", signif(pval, 3)),
           fontface = "italic", size = 4) +
  labs(
    x = "",
    y = expression(Log[2]*" fold-enrichment"),
    title = "a"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    panel.grid.major.x = element_blank(),
    legend.position = "none",
  )

log2fc_plot
```




```
# get overall genome coverage enrichment - use wilcox_summary table which includes NA as a non detectio
overall_cov_tbl = wilcox_summary %>%
  group_by(virus) %>%
```

```r
  summarize(cov_change_mean = mean(percent_enrichment), .groups = "drop") # change to depth if you want

mean(overall_cov_tbl$cov_change_mean)
```

```
## [1] 9.89
```

```r
round(quantile(overall_cov_tbl$cov_change_mean, 0.25, na.rm = TRUE), 2)
```

```
##  25%
## 4.54
```

```r
round(quantile(overall_cov_tbl$cov_change_mean, 0.75, na.rm = TRUE), 2)
```

```
##   75%
## 11.84
```

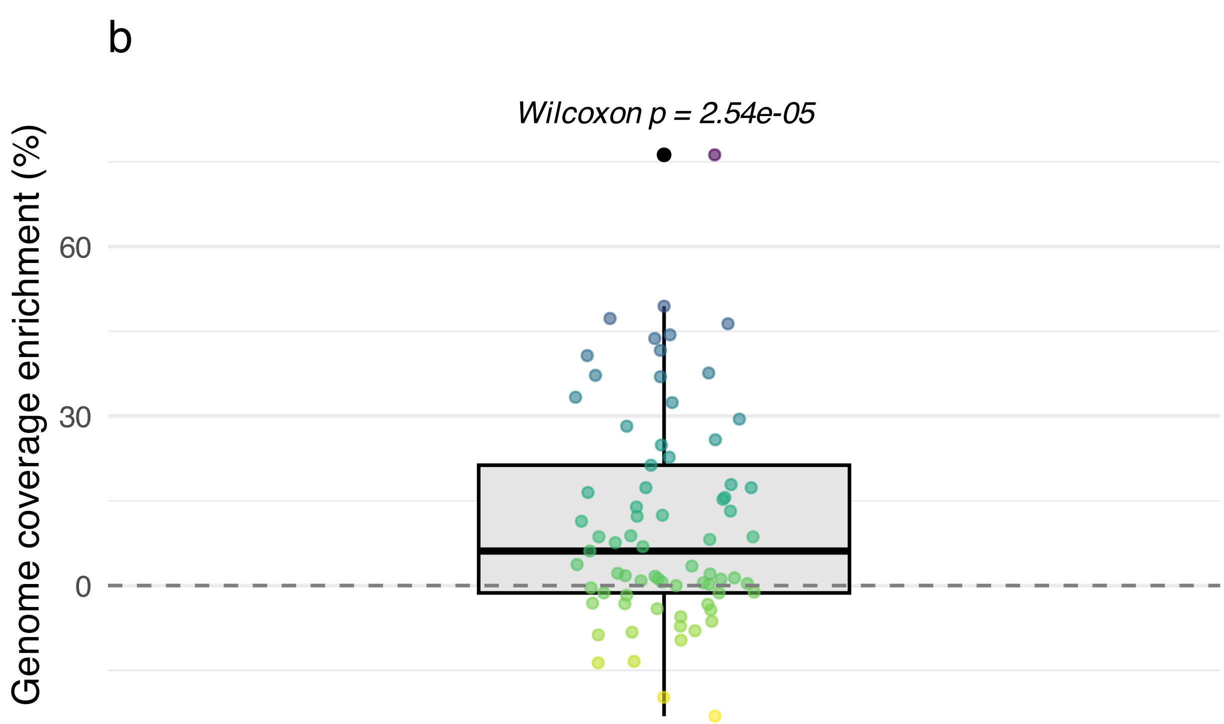
```r
# Is the different in coverage values significant?
wilcox.test(
  paired_tbl$breadth_1x_NGS,
  paired_tbl$breadth_1x_MSSPE,
  paired = TRUE,
  alternative = "two.sided"
)
```

```
##
##  Wilcoxon signed rank test with continuity correction
##
## data:  paired_tbl$breadth_1x_NGS and paired_tbl$breadth_1x_MSSPE
## V = 563, p-value = 2.536e-05
## alternative hypothesis: true location shift is not equal to 0
```

```r
# Paired Wilcoxon test across all samples
wilcox_res_cov <- wilcox.test(
  paired_tbl$breadth_1x_NGS,
  paired_tbl$breadth_1x_MSSPE,
  paired = TRUE,
  alternative = "two.sided"
)

pval_cov <- wilcox_res_cov$p.value

# summarize for boxplot
overall_cov_tbl_2 = cov_summary %>%
  group_by(virus, prep) %>%
  summarize(cov_change_mean = mean(breadth_1x, na.rm = TRUE), .groups = "drop")

# Boxplot of mean_nt_rpm by library_prep
cov_plot = ggplot(paired_tbl, aes(x = "", y = enrichment_breadth)) +
  geom_boxplot(width = 0.4, fill = "gray90", color = "black") +
  geom_jitter(width = 0.1, alpha = 0.6, size = 1.5, aes(color = enrichment_breadth)) +
```

```r
  scale_color_viridis(option = "viridis", direction = -1) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
  annotate("text", x = 1, y = max(paired_tbl$enrichment_breadth, na.rm = TRUE) * 1.1,
           label = paste0("Wilcoxon p = ", signif(pval_cov, 3)),
           fontface = "italic", size = 4) +
  labs(
    x = "",
    y = "Genome coverage enrichment (%)",
    title = "b"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    panel.grid.major.x = element_blank(),
    legend.position = "none",
  )

cov_plot
```

```
## Warning: Removed 23 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

## Warning: Removed 23 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

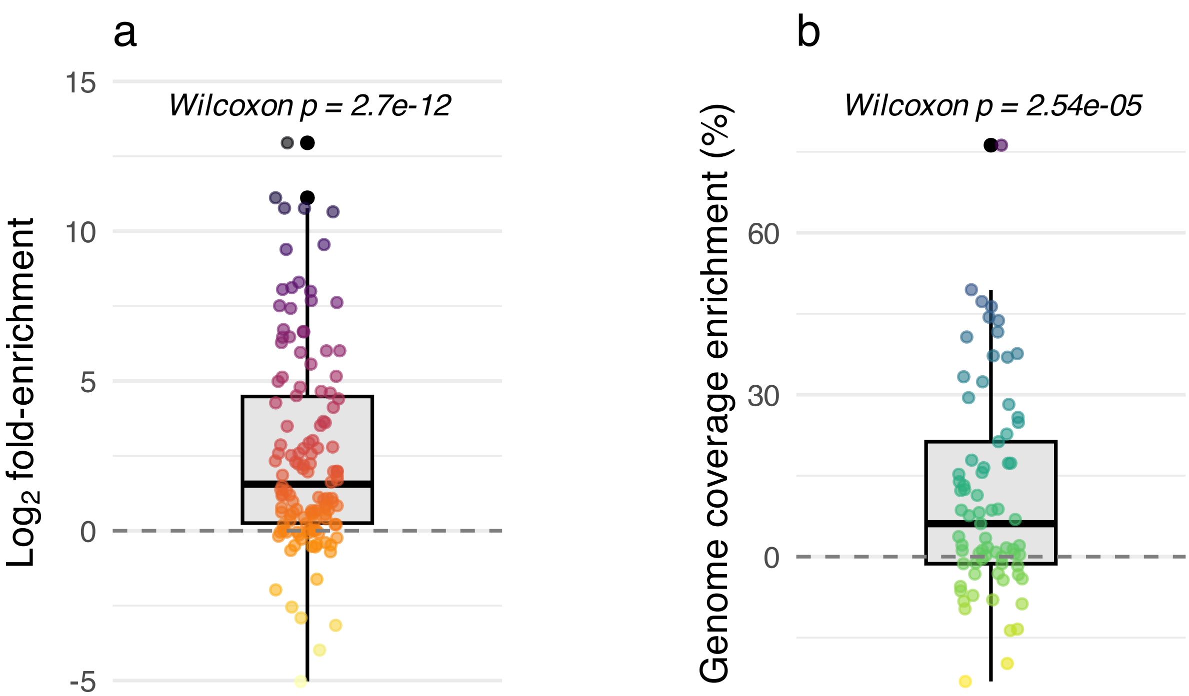



```r
horizontal_space <- plot_spacer() + plot_layout(widths = c(0.05))

combined_boxplots = (log2fc_plot | horizontal_space | cov_plot) +
  plot_layout(ncol = 3, widths = c(1.5, 0.5, 1.5))
```

```
combined_boxplots
```

```
## Warning: Removed 23 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
## Warning: Removed 23 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



Comparing Likelihood of detecting targets by Lib Prep or Sample Type

```
library(purrr)

# filter to targets
filter2targets = all_viruses %>%
  filter(str_detect(name_group, regex(paste0("\\b(", paste(name_targets, collapse="|"), ")\\b"), ignore_
  filter(name_group != "PToV*")  # remove torovirus

# get tables to create matrix
all_targets = unique(filter2targets$name_group)
all_samples = unique(filter2targets$sample_name)
all_preps = unique(filter2targets$library_prep)
all_types = unique(filter2targets$sample_type)

# create library prep matrix + detected column
expanded <- expand_grid(
  sample_name = all_samples,
  library_prep = all_preps,
  name_group = all_targets
) %>%
  left_join(filter2targets %>% mutate(detected = 1),
```

```r
            by = c("sample_name", "library_prep", "name_group")) %>%
  mutate(detected = ifelse(is.na(detected), 0, detected))

# do mcnemars test
results_libprep <- expanded %>%
  group_by(name_group, sample_name) %>%
  summarise(
    NGS = detected[library_prep == "NGS"],
    MSSPE = detected[library_prep == "MSSPE"],
    .groups = "drop"
  ) %>%
  group_by(name_group) %>%
  summarise(
    mcnemar = list(mcnemar.test(table(NGS, MSSPE))),
    p_value = mcnemar[[1]]$p.value,
    # Compute discordant counts
    b = sum(NGS == 0 & MSSPE == 1),  # detected by MSSPE only
    c = sum(NGS == 1 & MSSPE == 0),  # detected by NGS only

    # Odds ratio (b/c)
    odds_ratio = ifelse(c == 0, NA, b / c),
    log_or = log(odds_ratio),
    se_log_or = sqrt(1 / b + 1 / c),
    ci_lower = exp(log_or - 1.96 * se_log_or),
    ci_upper = exp(log_or + 1.96 * se_log_or),
    .groups = "drop"
  )
```

```
## Warning: Returning more (or less) than 1 row per `summarise()` group was deprecated in
## dplyr 1.1.0.
## i Please use `reframe()` instead.
## i When switching from `summarise()` to `reframe()`, remember that `reframe()`
##   always returns an ungrouped data frame and adjust accordingly.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

Overall Odds Ratio

```r
# Sum discordant counts across all viruses
overall_b <- sum(results_libprep$b, na.rm = TRUE)  # MSSPE only
overall_c <- sum(results_libprep$c, na.rm = TRUE)  # NGS only

# Construct 2x2 table for McNemar
overall_tbl <- matrix(
  c(NA, overall_b, overall_c, NA),  # we only need discordant counts for OR/p
  nrow = 2,
  byrow = TRUE,
  dimnames = list(NGS = c("0","1"), MSSPE = c("0","1"))
)

# Run McNemar test
# Note: mcnemar.test expects all 4 cells; can set concordant counts to 0
overall_tbl[1,1] <- 0  # neither detected
```

```r
overall_tbl[2,2] <- 0  # both detected

overall_test <- mcnemar.test(overall_tbl)

# Overall odds ratio
overall_or <- ifelse(overall_c > 0, overall_b / overall_c, NA)
log_or <- log(overall_or)
se_log_or <- sqrt(1/overall_b + 1/overall_c)
ci_lower <- exp(log_or - 1.96 * se_log_or)
ci_upper <- exp(log_or + 1.96 * se_log_or)

overall_results <- tibble(
  odds_ratio = overall_or,
  log_or = log_or,
  se_log_or = se_log_or,
  ci_lower = ci_lower,
  ci_upper = ci_upper,
  p_value = overall_test$p.value
)

overall_results
```

```
## # A tibble: 1 x 6
##   odds_ratio log_or se_log_or ci_lower ci_upper      p_value
##        <dbl>  <dbl>     <dbl>    <dbl>    <dbl>        <dbl>
## 1       1.59  0.467    0.0837     1.35     1.88 0.0000000235
```

Genus Odds Ratios